

REEV CONCEPT

Range-Extended Electric Vehicle

India-Focused Architecture Concept · INGLO Platform Derivation

Concept document · August 2026 · Pre-feasibility stage · Not for regulatory submission

1. Executive Summary

This document introduces a Range-Extended Electric Vehicle (REEV) concept designed specifically for Indian driving and infrastructure conditions. The concept is derived from the Mahindra XEV 9S INGLO BEV platform and retains its 400 V architecture, rear electric drive unit, and proven body structure. The combustion engine never drives the wheels — traction is always electric.

The core proposition: a 40/50 kWh battery pack handles daily city commuting without the engine starting, while a dedicated 105 kW petrol generator provides unlimited range on any highway in India. The buyer gets a city-capable EV that never needs a charger on a long trip.

Item	RWD Variant	AWD Variant
Battery (usable)	40 kWh	50 kWh
Peak battery power	187 kW (4.67C)	263 kW (5.26C)
Traction architecture	Rear EDU — always electric	Dual EDU — always electric
Generator output	105 kW HV-DC peak	105 kW HV-DC peak
Fuel tank	42 L petrol	42 L petrol
Programme kerb target	2,300 kg	2,500 kg
MIDC EV range (calibrated)	~340 km	~408 km
Real-world EV range	~200–260 km	~240–310 km
Total range	Unlimited — petrol backstop	Unlimited — petrol backstop
HV bus voltage	400–465 V nominal	400–465 V nominal

2. The Problem This Solves

Three distinct customer anxieties prevent BEV adoption in India that this architecture resolves simultaneously:

- **Range and charging anxiety** Charging infrastructure is too sparse for highway use. Less than 20% of Indian highways have DC fast chargers at the density required for a pure BEV. The REEV's generator eliminates this concern entirely — petrol is available at over 80,000 fuel stations across India.
- **Petrol running cost** Running on petrol at 8–10 km/l, a premium SUV costs ₹10–13/km in city conditions. The same REEV route on grid electricity costs approximately ₹2/km — one-fifth of the petrol bill for the identical commute.
- **Diesel maintenance and 10-year ban** BS6-II diesel SUVs carry DPF replacement costs of ₹60,000–₹1,20,000 and AdBlue consumption of ₹2,000–₹4,000 per service. SC and NGT orders prohibit diesel vehicles older than 10 years in Delhi NCR and expanding metro zones.
- **Strong hybrid — cost without the escape route** Strong hybrids at 40% GST carry the same tax as a full ICE vehicle. With the proposed 18% REEV classification, this concept would retail at a lower on-road price than an equivalent strong hybrid while delivering genuine daily EV range.

3. Platform Derivation — Mahindra INGLO

The concept is built on the XEV 9S INGLO skateboard. The INGLO platform carries most of its investment in the body, suspension, electronics, and HV architecture. A REEV derivative touches only two areas: the energy storage and the space behind the firewall.

System	INGLO / XEV 9S baseline	REEV modification
Body structure (BIW)	Retained	Purpose-built REEV floor for shorter pack
Front suspension	Retained	Unchanged
Rear EDU (traction motor)	Retained — drives wheels	Retained — still drives wheels in RWD
AWD front EDU	Present in AWD variant	Retained; housing carried in RWD body (structural)
400 V HV architecture	Retained	Same 108-series cell string, same bus
79 kWh Blade 1.0 LFP pack	Replaced	40/50 kWh Short Blade 2.0 high-power phosphate
Engine / generator / fuel	Not present	Added: 1.5T TGD _i + PMSG + 42 L tank
Exhaust / after-treatment	Not present	Added

4. Battery Pack — Why 40/50 kWh, and Why These Cells

The pack sizes are not chosen to maximise range. They are set by peak power demand at a healthy C-rate. The motor calibration requires 187 kW DC from the RWD pack and 263 kW from the AWD pack.

Variant	Usable capacity	Peak power demand	C-rate	Cell Ah (108s 1p)	Bus voltage
RWD	40 kWh	187 kW	4.67C	~97.5 Ah	~410 V nominal
AWD	50 kWh	263 kW	5.26C	~121.8 Ah	~410 V nominal
XEV 9S BEV (reference)	79 kWh	187 kW	2.37C	~192 Ah	~410 V nominal

The 79 kWh INGLO Blade 1.0 LFP cell runs at a comfortable 2.37C. At 40 kWh the C-rate nearly doubles — the Blade 1.0 was not characterised for this duty. The BYD Short Blade 2.0 high-power phosphate variant is designed for PHEV/REEV shallow cycling, high pulse power, and repeated generator charging.

The 40/50 kWh pair is the smallest round-kWh split at which both variants stay at or below 5.3C with minimal asymmetry. Both use a 108-series 1-parallel string, maintaining the 400 V bus without any step-up converter. Certified MIDC EV range — ~340 km RWD, ~408 km AWD — comfortably exceeds the proposed 270 km policy threshold. This headroom is a consequence of correct pack sizing, not a design target.

5. Engine-Generator System

The 1.5-litre turbocharged petrol engine (mStallion TGD_i class, ~105 kW crank output) drives a PMSG through a single fixed-ratio coupling. The rectified HV-DC output feeds directly onto the 400 V bus. The engine has no mechanical path to the wheels at any point.

Band	HV-DC output	Engine RPM	Brake thermal efficiency	Typical use
1	30 kW	1,500	38.5–40.0%	Turtle recovery / minimum sustaining
2	40 kW	1,800	37.5–39.0%	Light highway
3	55 kW	2,300	36.5–38.0%	Normal highway

4	70 kW	2,800	35.0–37.0%	Sustained high load
5	90 kW	3,500	33.5–35.5%	High demand / hills
6	105 kW	4,000	31.5–34.0%	Maximum output

6. Energy Management — Control Law Summary

State	Condition	Generator	Battery role
S1 — Pure EV	SoC above mode start boundary	Off	Supplies all traction and auxiliaries
S2 — Battery protection	SoC healthy but capability limited	Off	Torque reduced; no generator start for acceleration
S3 — Charge sustaining	SoC at mode start boundary	Running in smallest sufficient band	Covers transients; surplus recharges battery
S4 — Generator limited	Battery cannot sustain traction deficit safely	Running; covers auxiliaries + traction	Discharge limit zero; accepts generator surplus only
S5 — Turtle	Battery at Turtle floor, no fuel	Off	Emergency mobility; 60 km/h ceiling
S6 — Refuel recovery	Fuel confirmed from S5	Restarting at 30–40 kW	Energy rebuilt before performance restored

Three driver-selectable modes — Economy (15%/30%), Sport (20%/35%), Hills/Terrain (20%/35% with longer downshift dwell) — adjust generator boundaries and ramp rates. Sport demotes to Economy at 9.5% SoC. Hills/Terrain holds its band-hunting character down to 8%.

7. Mass Budget

XEV 9S reference: 2,240 kg with 79 kWh pack. All values are planning ranges, not weighed results:

Item	RWD Δ kg	AWD Δ kg
Remove: 79 kWh Blade 1.0 LFP pack	–550 to –560	–550 to –560
Add: 40 kWh Short Blade 2.0 RWD pack	+240 to +270	—
Add: 50 kWh Short Blade 2.0 AWD pack	—	+295 to +330
Add: Engine + fluids + PMSG + coupling + rectifier + intake	+230 to +290	+230 to +290
Add: Exhaust + after-treatment + cooling + NVH shields	+54 to +85	+54 to +85
Add: 42 L fuel tank + pump + lines + vapour system	+40 to +50	+40 to +50
Add: Thermal / structural / NVH integration	+15 to +30	+15 to +30
Add: AWD front 70 kW EDU + subframe + HV hardware	—	+145 to +195
BIW / floor optimisation opportunity	–20 to –70	–20 to –70
Planning range vs XEV 9S	~2,170 to ~2,330 kg	~2,360 to ~2,545 kg
Programme kerb target	2,300 kg	2,500 kg

8. Running Costs vs Alternatives

Road-load model at 2,300 kg RWD, 100 km/h highway, normal load. REEV assumes 95% grid charging in city. Fuel prices: petrol ₹104/L, diesel ₹92/L, electricity ₹8/kWh.

Vehicle	City ₹/km	Highway ₹/km	10-yr fuel cost (10,000 km/yr)
REEV RWD — city on grid	~₹2.0	~₹4.5 (generator)	~₹3.3 lakh
XEV 9S BEV (79 kWh)	~₹2.0	~₹3.8	~₹3.0 lakh
2.2 mHawk diesel AT	~₹8.2	~₹6.2	~₹7.2 lakh
2.0 mStallion petrol AT	~₹13.7	~₹10.6	~₹12.2 lakh
Illustrative 185 hp HEV	~₹5.6	~₹6.1	~₹5.9 lakh

9. Policy Proposal — GST Classification

A capability-based GST class defined by measurable, independently verifiable criteria. Current GST on an over-4-metre petrol SUV above 1,500 cc is 40%:

Class	Proposed GST	Generator displacement	Certified MIDC all-electric range
Above 4 metres	18%	Above 1,000 cc and below 1,500 cc	At least 270 km
Sub-4 metres	9%	500 cc to below 1,000 cc	At least 210 km
At or above 1,500 cc	40%	1,500 cc and above	No preferential rate

The fiscal case: GST forgone is ~₹0.86 lakh net delta per vehicle (₹4.73L forgone minus ₹3.87L still collected at 18%). Crude import saving recovers that net delta in under 4 years for a typical user. At 1 lakh vehicles the ₹860 crore net delta is recovered from ₹286 crore/year in avoided crude imports in approximately 3 years.

10. Competitive Context

Vehicle	Architecture	Pack	Key distinction vs this concept
Li Auto L6 (China)	Series REEV	52.3 kWh	Premium estate; no mechanical wheel path
Leapmotor C10 REEV (Europe)	Series REEV	28.4 kWh	B-segment; smaller pack; lower price point
Aito M7 (China)	Series REEV (Huawei Driveone)	~40 kWh	Premium SUV; HarmonyOS integration
MG ADAPT (India, Jul 2026)	DHT — parallel hybrid + REEV mode	20.5 kWh PHEV	Multi-mode; REEV deprioritised; 400V capable
This concept	Dedicated series REEV only	40 / 50 kWh	INGLO derivation; India-specific control law; always-electric wheels

11. Open Questions and Next Steps

- **Pack and cell** Cell selection, pack topology, voltage and current maps, pulse durability, thermal analysis — BYD Short Blade 2.0 is the candidate but supplier datasheet data is required.
- **Engine-generator** Engine-generator dyno maps for the specific mStallion TGD_i + PMSG unit. Band BTE values are provisional abstractions.
- **Road-load constants** Coast-down testing to confirm CdA, Crr, and kerb mass on a production-representative vehicle.
- **Regenerative braking** Brake-by-wire hardware design, ABS/ESC integration, and homologation for the 70% regen recovery assumption.

- **ARAI certification** ARAI OVC-HEV procedure agreement — test family, variant selection, and break-off criterion require regulatory alignment.

This document is a concept introduction for discussion purposes. All figures are derived from the road-load model and design specification. No vehicle has been built, tested, or homologated.